MOTION DOWN THE INCLINE

An object moving down a frictionless inclined plane is an example of uniformly accelerated motion. The object accelerates due to the action of the earth's gravity.

Ignoring friction, the acceleration component of the cart down the incline can be shown to be ***a = g sinθ***, where θ is the angle of the incline above the horizontal.

If we know the height h and length L of the track then we know that sinθ = h/L

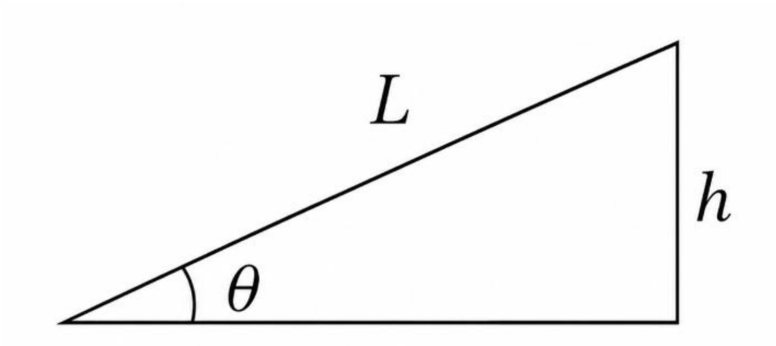

Figure 3.1. Relation of θ, L and h

Therefore, the acceleration of the object moving down a frictionless inclined plane is ***a = g(h/L)***.

SIGNIFICANT FORMULAS

SINE RELATIONSHIP

$$sin\theta = \frac{h}{L}$$

THEORETICAL ACCELERATION

$$a_{theo} = g\ sin\theta$$

or

$$a_{theo} = g\left(\frac{h}{L}\right)$$

PERCENT ERROR FOR ACCELERATION

$$\%error_a = \frac{|a_{theo} - a_{exp}|}{a_{theo}} \times 100$$

PERCENT ERROR FOR GRAVITATIONAL ACCELERATION

$$\%error_g = \frac{|g_{theo} - g_{exp}|}{g_{theo}} \times 100$$